

Experiment :- 2

AIM :- Explain the SDLC model used in the minor project “BizFlow - Inventory and Billing Management System” and why you used that specific Model.

For the project BizFlow - Inventory and Billing Management System, the Agile Model was used because the project required continuous development, testing, feature enhancements, and user feedback throughout the development process.

SDLC Model Used in Project :-

Agile Model

The Agile Model is a modern SDLC approach in which software is developed in small parts called iterations or sprints.

Each module is developed, tested, and improved continuously based on feedback and changing business requirements.

Why Agile Model was Selected :-

The following reasons explain why Agile Model was chosen for the project:

i) Continuous Improvement

The project required frequent updates and improvements in inventory management, billing functionality, and analytics features.

ii) Easy Testing

Every module could be tested separately after development to ensure proper functionality.

iii) Flexible Development

New features and changes could be added easily during the development process.

iv) Faster Development

The project was divided into smaller modules, making development and deployment faster.

v) Better User Experience

Continuous feedback helped improve usability and overall system performance.

Phases of Agile Model Used in Project :-

Phase 1: Requirement Gathering

In this phase, all project requirements were collected.

Requirements Identified:

- User Authentication
- Product Catalog Management
- Inventory Tracking
- Bill Generation
- Invoice PDF Generation
- Sales Analytics Dashboard
- Bill History Management

Outcome:

A clear understanding of project objectives and business requirements was created.

Phase 2: Planning

In this phase, planning for software tools, technologies, modules, and development timeline was completed.

Tools and Technologies Planned:

- React.js
- Vite
- JavaScript
- Supabase
- PostgreSQL
- Tailwind CSS
- PDF Generation Library

Outcome:

Project roadmap and module planning were prepared.

Phase 3: System Design

The system architecture and module design were prepared.

Designed Components:

- Authentication Module
- Product Catalog Module
- Inventory Management Module
- Billing Module
- Invoice Generation Module
- Analytics Dashboard Module

Outcome:

The structure of the complete system was finalized.

Phase 4: Development

Coding and implementation of modules were done.

Development Activities:

- Product management functionality
- Inventory tracking implementation
- Bill generation system
- Invoice PDF generation
- Sales analytics development
- Database integration using Supabase

Outcome:

Working modules of the project were developed.

Phase 5: Testing

Each module was tested separately.

Types of Testing Performed:

- Unit Testing
- Integration Testing

- System Testing

Errors Fixed:

- Inventory calculation issues
- Billing calculation errors
- Database synchronization problems
- User interface bugs

Outcome:

The system performance and reliability improved significantly.

Phase 6: Deployment

After successful testing, the project was deployed and executed.

Deployment Activities:

- Application deployment
- Database configuration
- User verification
- Functional testing in production environment

Outcome:

The project became operational and ready for practical use.

Phase 7: Maintenance

After deployment, improvements and updates were performed.

Maintenance Activities:

- Fixing software bugs
- Improving application performance
- Adding new features
- Updating database structures

Outcome:

The project performance remained stable, reliable, and efficient.

Advantages of Using Agile Model in This Project :-

- i) Faster Development

Modules were developed and delivered quickly.

ii) Easy Error Detection

Errors were identified and fixed during each development cycle.

iii) Continuous Feedback

Regular improvements were possible based on user requirements.

iv) Flexible System

New features could be added without affecting existing functionality.

v) Better Project Management

Tasks and modules were organized efficiently throughout development.